

Raspberry Pi–Based Infrared Temperature Monitoring and Safety Interlock for a Laboratory Plasma Reactor

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1 Introduction

Modern plasma reactors often operate at high voltage and high frequency in confined geometries, where direct contact temperature measurements at critical components are technically challenging and may perturb the process conditions or the mechanical integrity of the reactor. In the laboratory system considered here, the plasma discharge is confined in a glass tube that is sealed at one end by a thin ceramic end cap. Experimental experience indicates that this ceramic seal becomes electrically damaged once its temperature exceeds approximately 200 °C, at which point the plasma discharge collapses and the experiment has to be stopped.

Non-contact infrared (IR) thermography is widely used to obtain spatially and temporally resolved temperature information in chemical reactors and catalytic systems, because it can provide surface temperature maps without disturbing the process.[1] Several recent studies have demonstrated that IR cameras and pyrometers can be employed to monitor heterogeneous reactions, microwave or plasmatron heating, and other high-temperature processes in operando.[2] However, the accuracy of such non-contact temperature measurements depends strongly on the spectral emissivity of the observed materials, which in turn depends on surface condition, wavelength, and temperature.[3–6] For ceramic and glass-ceramic materials, emissivity values are often

high in broad wavelength ranges but can exhibit significant spectral variation, so appropriate calibration or conservative safety margins are required.

In the present reactor configuration, the temperature of the ceramic seal cannot be measured directly with conventional contact sensors such as thermocouples or resistance temperature detectors, because there is no straightforward way to embed a sensor at the seal without affecting the plasma, the vacuum integrity, or the mechanical strength of the component. At the same time, there is currently no integrated monitoring and shutdown mechanism that would reduce or disable the high-voltage power supply when the ceramic approaches its critical temperature. This poses a risk of repeated component damage and unplanned interruptions of the experimental campaign.

In parallel, low-cost embedded computing platforms such as the Raspberry Pi have become a common basis for small-scale monitoring and control tasks in research laboratories. There exists an active ecosystem of thermal camera modules (e.g. MLX90640, AMG8833) and open-source software that enable IR image acquisition, temperature field visualisation, and basic processing directly on the Raspberry Pi.^[7–9] These developments make it feasible to implement a compact non-contact temperature monitoring and shutdown system tailored to the specific geometry and constraints of the plasma reactor.

The problem addressed in this project can therefore be summarised as follows: the laboratory plasma reactor currently lacks a non-contact, automated temperature monitoring and shutdown system that protects the ceramic end seal from overheating, despite a known approximate destruction threshold around 200 °C and the impracticability of inserting conventional contact sensors at the critical location.

2 Goal of the Project

The overall goal of the project is to design, implement and experimentally validate a Raspberry-Pi-based infrared temperature monitoring and overtemperature shutdown system for a laboratory plasma reactor with a ceramic end seal.

More concretely, the project aims to:

- continuously measure the apparent temperature of the ceramic seal using an IR thermal camera or IR array sensor connected to a Raspberry Pi;
- implement a region-of-interest (ROI) algorithm that focuses specifically on the ceramic area in the thermal image and extracts a robust temperature indicator for this region (e.g. maximum or high-percentile temperature);
- design and implement a multi-level temperature evaluation logic with warning and trip thresholds, including hysteresis and minimum-duration criteria, in order to detect unsafe heating before the estimated 200 °C limit is reached;
- integrate the Raspberry Pi with the plasma power supply via an RS-232 communication interface and, optionally, via a simple digital output or relay, so that the high-voltage output is automatically disabled when an overtemperature trip is detected;

- validate the behaviour of the complete system in experiments with the real reactor, and document its performance, limitations and potential extensions towards more industrial-style safety interlocks.[10, 11]

3 Expected Contributions

The project is expected to provide several concrete technical and methodological contributions:

3.1 Prototype of a Temperature-Based Safety Interlock

The first contribution is an end-to-end prototype of a temperature-based safety interlock. The system will cover the complete chain from non-contact sensor to shutdown action:

IR thermal sensor $\rightarrow$ *Raspberry Pi (monitoring and logic)* $\rightarrow$ *RS-232 / relay output*
 $\rightarrow$ *plasma power supply*.

The logic will be inspired by established patterns from industrial overtemperature limit controllers, such as latching trip outputs, multi-level alarms and remote reset functionality.[10, 11] The design will be documented in a way that can be reused or extended for similar laboratory-scale systems.

3.2 ROI-Based IR Monitoring of a Ceramic Reactor Seal

The second contribution is an image-processing pipeline for ROI-based IR monitoring of the ceramic seal. This includes:

- defining a stable ROI that encompasses the ceramic end cap in the thermal image;
- computing representative temperatures within the ROI (maximum, mean, and selected quantiles);
- discussing the impact of emissivity assumptions, viewing geometry and plasma emission on the measured values, in light of existing literature on ceramic emissivities and contactless measurement uncertainty.[2, 3, 5]

Although the primary objective is protection against overheating rather than precise metrology, the project will explicitly document how conservative safety thresholds are derived from the available emissivity information and calibration steps.

3.3 Embedded Software Framework for Thermal Monitoring

The third contribution is a compact software framework for thermal monitoring on the Raspberry Pi. The framework will be implemented in Python and will provide:

- modules for IR image acquisition from the selected thermal camera or array sensor;
- a state-machine-based implementation of the alarm and trip logic (including hysteresis and minimum time-above-threshold criteria);

- RS-232 communication routines to send control commands to the plasma power supply;
- optional general-purpose input/output (GPIO) control for a relay output, which can emulate a hardware interlock or *enable* signal.

The design will make use of existing open-source examples and tutorials for interfacing thermal cameras with Raspberry Pi,^[7–9] but will extend them by integrating a structured protection logic and power-supply control.

3.4 Experimental Case Study on Plasma Reactor Protection

Finally, the project will provide an experimental case study that characterises the temperature profile of the ceramic seal under different plasma operating conditions, and demonstrates the effectiveness of the shutdown mechanism. The resulting data can be used to:

- illustrate typical heating dynamics of the ceramic seal over time;
- verify that warning and trip thresholds are triggered as intended;
- confirm that, under realistic operating conditions, the ceramic temperature can be kept below a chosen safety threshold (e.g. 180 °C) by the automated shutdown.

4 Work Plan

The planned duration of the project is approximately three months. The work can be structured into three phases.

4.1 Phase 1 (Weeks 1–3): Requirements, Background Study and Hardware Setup

Tasks.

- Finalise the system requirements:
 - target sampling rate (e.g. 2–10 Hz) and expected temperature range (e.g. 20–250 °C);
 - warning and trip thresholds including safety margins (e.g. trip at 180 °C for an estimated 200 °C limit);
 - required interfaces to the plasma power supply (RS-232 protocol and, optionally, digital input for interlock).
- Conduct a focused literature and application review on:
 - non-contact IR temperature measurement and IR thermography in reactors, highlighting examples of catalytic and plasma systems;^[1, 2]
 - emissivity and its influence on measurement uncertainty for ceramic and glass-ceramic materials;^[3–6]
 - typical overtemperature safety interlock architectures and limit controllers with latching outputs.^[10, 11]
- Select and assemble the hardware:

- choose a suitable IR thermal camera or array module that can be interfaced with Raspberry Pi (e.g. MLX90640 or AMG8833);[7–9]
- set up the Raspberry Pi, power supply and RS-232 converter, and verify basic serial communication with the plasma power supply.

Deliverables.

- A concise background survey (approximately two to three pages) summarising the state of the art.
- A confirmed hardware list and a basic demonstration that the Raspberry Pi can acquire IR data from the sensor and communicate with the plasma power supply.

4.2 Phase 2 (Weeks 4–7): Software Development and Temperature Logic

Tasks.

- **IR data acquisition:**
 - implement a Python module to read temperature frames or arrays from the thermal camera;
 - store raw frames for offline analysis and verification.
- **ROI and temperature extraction:**
 - define the ROI corresponding to the ceramic end seal, either via fixed image coordinates or a configuration file;
 - compute for each frame the ROI maximum temperature, mean temperature and a selected quantile (e.g. 95 %).
- **Alarm and trip state machine:**
 - implement a multi-level logic with states such as *NORMAL*, *WARNING* and *TRIPPED*;
 - incorporate warning and trip thresholds, hysteresis bands and minimum-duration criteria for threshold violations;
 - implement a latching trip that requires an explicit reset command before operation can resume, in analogy to industrial limit controllers.[10, 11]
- **RS-232 integration:**
 - implement functions that send appropriate commands (e.g. “HV off” or “set voltage to zero”) to the plasma power supply when an overtemperature trip occurs;
 - implement acknowledgement handling and basic error reporting.
- **Optional GPIO + relay output:**
 - design and test a simple relay output driven by a Raspberry Pi GPIO pin, which can emulate an interlock or *enable* contact for the power supply.
- **User interface and logging:**

- implement a simple command-line interface or minimal graphical interface to display the current ROI temperature, system state and recent history;
- log temperature and events (warnings, trips, resets) to files with time stamps for later analysis.

Deliverables.

- A working prototype that can be exercised with simulated or recorded temperature data, demonstrating the correct alarm and trip behaviour.
- Source code with basic documentation (README and in-code comments).

4.3 Phase 3 (Weeks 8–12): Integration, Experiments and Evaluation

Tasks.

- **Integration with the plasma reactor:**
 - mount the IR camera in a mechanically stable configuration with a clear view of the ceramic seal;
 - configure emissivity settings and, if feasible, perform a simple calibration using external heating and a reference sensor, guided by available emissivity data.[3–6]
- **Experimental campaign:**
 - operate the plasma reactor at different power levels and duty cycles while recording the seal temperature and system state;
 - verify that warning messages, alarm states and trip events are triggered as expected;
 - confirm that the automated shutdown prevents the ceramic from exceeding the chosen safety threshold in representative operating scenarios.
- **Performance and limitation analysis:**
 - analyse the uncertainty of the non-contact temperature measurements in light of emissivity variation, plasma radiation and viewing through glass;[2]
 - compare the developed prototype conceptually with industrial overtemperature protection architectures based on dedicated safety limit controllers and certified safety relays, and clearly delineate the scope and limitations of the prototype.[10, 11]
- **Documentation:**
 - compile a final written report that documents the system architecture, algorithms, implementation details and experimental results;
 - outline possible future extensions, such as dual-channel sensing, independent hardware interlocks, higher-resolution cameras or integration into a broader laboratory control framework.

Deliverables.

- A functioning prototype system demonstrated on the real plasma reactor.

- A final report and a short presentation summarising the results.

References

- [1] Maurer, F., Becker, P., Förster, A., *et al.*: Spatiotemporal investigation of the temperature and structure of a fixed-bed reactor under dynamic operation. *Ind. Eng. Chem. Res.* **60**(12), 4545–4560 (2021) <https://doi.org/10.1021/acs.iecr.0c05798>
- [2] Fresneda-Cruz, A., Arenillas, M., Arenillas, J., *et al.*: Experimental dielectric properties and temperature measurement uncertainty in contactless thermographic cameras and infrared pyrometers. *Processes* **13**(3), 774 (2025) <https://doi.org/10.3390/processes13030774>
- [3] Bauer, W., Moldenhauer, A.: Emissivities of ceramics for temperature measurements. In: *Proc. SPIE*, vol. 5405 (2004). <https://doi.org/10.1117/12.538739> . <https://doi.org/10.1117/12.538739>
- [4] Yang, T.C.K., Yen, S.K., Shieh, T.H.: In-situ measurement of emissivities of ceramic–polymer composites. *J. Am. Ceram. Soc.* **80**(2), 401–408 (1997) <https://doi.org/10.1111/j.1151-2916.1997.tb02804.x>
- [5] Fan, M., Li, W., Zhang, L., *et al.*: Measurement of thermal properties and numerical simulation of glass–ceramic materials. *Silicon* **15**, 2489–2502 (2023) <https://doi.org/10.1007/s12633-022-01907-0>
- [6] AG, T.: Table of emissivity of various surfaces. Technical note (2016). https://www.transmetra.ch/images/transmetra_pdf/publikationen_literatur/pyrometrie-thermografie/emissivity_table.pdf
- [7] MakersPortal: High resolution thermal camera with Raspberry Pi and MLX90640 (2020). <https://makersportal.com/blog/2020/6/8/high-resolution-thermal-camera-with-raspberry-pi-and-mlx90640>
- [8] How2Electronics: DIY thermal imaging camera with MLX90640 Raspberry Pi (2024). <https://how2electronics.com/diy-thermal-imaging-camera-with-mlx90640-raspberry-pi/>
- [9] Sarraf, G.: Interfacing a thermal camera with Raspberry Pi: Complete setup guide. *ThinkRobotics Blog* (2025). <https://thinkrobotics.com/blogs/tutorials/interfacing-a-thermal-camera-with-raspberry-pi-complete-setup-guide>
- [10] Brainchild Electronic Co., L.: L41 safety limit controller. <https://www.brainchildtw.com/shop/selectseries-safety-limit-controller-l41-55>
- [11] Controller, M.: Brainchild l91 limit controller user manual. Technical manual. <https://www.intech.co.nz/wp-content/uploads/2019/09/>

